1. Miller, S.A. and Harley, J.B., 1999 & 2002. Zoology, 4th & 5th Edition (International). Singapore: McGraw Hill.
2. Hickman, C.P., Roberts, L.S. and Larson, A., 2004. Integrated Principles of Zoology, 11th Edition (International). Singapore: McGraw Hill.
3. Pechenik, J.A., 2000. Biology of Invertebrates, 4th Edition (International). Singapore: McGraw Hill.
4. Kent, G.C. and Miller, S., 2001. Comparative Anatomy of Vertebrates. New York: McGraw Hill.
5. Campbell, N.A., 2002. Biology Sixth Edition. Menlo Park, California: Benjamin/Cummings Publishing Company, Inc

Semester – III

Module Code:	Eng - 201
Module title:	**English – III (Technical Report Writing & Presentation Skills)**
Name of Scheme:	BS Chemistry (4 Years)
Semester :	3rd
Module Type:	Compulsory
Module Rating:	3 Credits

1. **Introduction of the Course:**
The main purpose of this course is to guide students about presentation skills including essay writing. The course will enable students to devise and follow "study systems" and equip them with the ability to think critically and adopt effective learning strategies.

2. **Course Objectives:**
The course aims to:
   1. Enhance language skills.
   2. Develop critical thinking.

3. **Course Contents:**
Presentation skills: Essay writing: Descriptive, narrative, discursive, argumentative.
Academic writing: How to write a proposal for research paper/term paper How to write a research paper/term paper (emphasis on style, content, language, form, clarity, consistency).
Technical Report writing Progress report writing
Note: Extensive reading is required for vocabulary building

4. **Teaching-learning Strategies**
   1. Lectures
   2. Group Discussion
   3. Laboratory work
   4. Seminar/ Workshop

5. **Learning Outcome:**
   1. Students are expected to get familiarized with the Basics of presentation skills.
   2. They will learn about the basic rules of paragraph writing and essay writing.

6. **Assessment Strategies**:
   1. Lecture Based Examination (Objective and Subjective)
   2. Assignments
   3. Class discussion
   4. Quiz
   5. Tests

7. **Recommended Readings**:
1. Kirszner. L.G., Mandell, S. R. Patterns of College Writing. 4th Ed. by St. Martin's Press.
2. Langan, J. 2004. College Writing Skills McGraw-Hill Higher Education.
3. Neulib, J., Cain, K. S., Ruffus, S., Scharton, M. (Editors). Reading. The Mercury Reader. A Custom Publication. Compiled by northern Illinois University. (A reader that will give students exposure to the best of twentieth century literature).
4. White, R. 1992. Writing. Advanced. Oxford Supplementary Skills. Third Impression (particularly suitable for discursive, descriptive, argumentative and report writing).

Module Code:	Chem - 201
Module title:	**Chemistry – III (Physical Chemistry)**
Name of Scheme:	BS Chemistry (4 Years)
Semester :	3rd
Module Type:	Foundation
Module Rating:	3 Credits

1. **Introduction of the Course:**
The course is organized to provide an adequate knowledge about basic concepts in Physical chemistry including thermodynamics, chemical kinetics etc.
2. **Course Objectives:**
The course is designed:
   1. To introduce students about the key concepts of physical chemistry
   2. To introduce about thermodynamics, chemical kinetics etc.

3. **Course Contents**
Theory
   1. **Chemical Thermodynamic**
   Equation of states, ideal and real gases, the vander waals equation for real gases, critical phenomena and critical constants.
   Extensive and intensive properties, molar heat capacities, second law of thermodynamics, concept of entropy, entropy change in reversible and irreversible process, entropy change for an ideal gas, entropy change due to mixing of ideal gases, effect of temperature and pressure on entropy, concept of free energy, effect of temperature and pressure on free energy, relationship between standard free energy and equilibrium constants.
   2. **Chemical Kinetics**
   Derivation of kinetics expression of zero order, first order, second order (with same and different concentrations), nuclear decay as first order reaction, derivations for determining rate constants and half life periods, measurement of order of the reaction with different methods, Arrhenius equation and determination of various Arrhenius parameters.
   3. **Solutions and Colloids**
   Physical properties of liquids, surface tension, viscosity, refractive index etc.
   Osmotic pressure and its measurements, abnormal colligative properties (association and dissociation of solutes), fractional distillation and concept of azeotrops, concept of colloids, classification of colloids, dialysis, electro-dialysis, sedimentation, precipitation, ultra filtration, emulsions and gels, tyndall cone effect.
   4. **Surface Chemistry**
   Interface, Adsorption, types of adsorption at liquid surface, adsorption isotherms (Freundlich and Langmuir), catalysis, and kinetics of enzyme catalysis.

4. **Teaching-learning Strategies**
   1. Lectures
   2. Group Discussion

   3. Laboratory work
   4. Seminar/ Workshop

3. Laboratory work
4. Seminar/ Workshop

5. **Learning Outcome:**
Students are expected to get familiarized with the with the basic concepts of Physical chemistry like chemical thermodynamics, solution chemistry and surface chemistry.

6. **Assessment Strategies**:
   1. Lecture Based Examination (Objective and Subjective)
   2. Assignments
   3. Class discussion
   4. Quiz
   5. Tests

7. **Recommended Readings:**

1. Adamson A. W. "Understanding Physical Chemistry" 3rd Ed., Benjamin Cummings Publishing Company Inc.
2. Akhtar M.N.& Ghulam Nabi, "Textbook of Physical Chemistry", ilmi Kutab Khana, Lahore. 3. Bhatti H.N. and K.Hussain, "Principles of Physical Chemistry"; Carwan Book House, Lahore.
3. Maron S.H. & B. Jerome, "Fundamentals of Physical Chemistry", Macruthan Publishing Co., Inc. New York. (Also published by National Book Foundation).
4. Atikins P.W.& M.J.Clugston, "Principles of Physical Chemistry" Pitman Publishing Company (1988).
5. Moore W.J. "Physical Chemistry", 5th Ed. Longmans Publishers.
6. Jones M. "Elements of Physical Chemistry" Addison-Sesky Publishing Company.
7. G.M.Barrow, International six Edition "Physical Chemistry".
8. IRA. N. Levine fourth edition "Physical Chemistry"
9. Alberty and Danials, "Physical Chemistry"
10. Castallon, "Physical Chemistry"
11. Laidler & Meiser "Physical Chemistry"
12. Friemental "Chemistry in Action"

16.

Module Code:	Chem - 202
Module title:	Chemistry – III (Physical Chemistry Lab)
Name of Scheme:	BS Chemistry (4 Years)
Semester :	3rd
Module Type:	Foundation
Module Rating:	1 Credits

1. **Introduction of the Course:**
The course is organized to provide an adequate knowledge about basic concepts in Physical chemistry including thermodynamics, chemical kinetics etc.
2. **Course Objectives:**
The course is designed:
   1. To introduce students about the key concepts of physical chemistry
   2. To introduce about thermodynamics, chemical kinetics etc.

3. **Course Contents**
<u>Lab</u>
   1. Preparation of standard molar, normal, molal and percentage solutions.
   2. Standardization of secondary standard acids and bases solutions by volumetric methods.
   3. Determination of surface tension, parachor and percentage composition by surface tension measurement.
   4. Determination of viscosity, rheochor and percentage composition by viscosity measurement.
   5. Determination of refractive index, molar refractivity and percentage composition by refractive index method.
   6. Conductometeric and potentiometric strong acid-base titrations using conductometer and pH meter respectively.

4. **Teaching-learning Strategies**

1. Lectures
2. Group Discussion
3. Laboratory work
4. Seminar/ Workshop

5. **Learning Outcome:**
Students are expected to get familiarized with the with the basic concepts of Physical chemistry like chemical thermodynamics, solution chemistry and surface chemistry.
6. **Assessment Strategies**:
   1. Lecture Based Examination (Objective and Subjective)
   2. Assignments
   3. Class discussion
   4. Quiz
   5. Tests
7. **Recommended Readings:**

1. Adamson A. W. "Understanding Physical Chemistry" 3rd Ed., Benjamin Cummings Publishing Company Inc.
2. Akhtar M.N.& Ghulam Nabi, "Textbook of Physical Chemistry", ilmi Kutab Khana, Lahore. 3. Bhatti H.N. and K.Hussain, "Principles of Physical Chemistry"; Carwan Book House, Lahore.
3. Maron S.H. & B. Jerome, "Fundamentals of Physical Chemistry", Macruthan Publishing Co., Inc. New York. (Also published by National Book Foundation).
4. Atikins P.W.& M.J.Clugston, "Principles of Physical Chemistry" Pitman Publishing Company (1988).
5. Moore W.J. "Physical Chemistry", 5th Ed. Longmans Publishers.
6. Jones M. "Elements of Physical Chemistry" Addison-Sesky Publishing Company.
7. G.M.Barrow, International six Edition "Physical Chemistry".
8. IRA. N. Levine fourth edition "Physical Chemistry"
9. Alberty and Danials, "Physical Chemistry"
10. Castallon, "Physical Chemistry"
11. Laidler & Meiser "Physical Chemistry"
12. Friemental "Chemistry in Action"

Module Code:	Bot - 201
Module title:	Botany – III (**Cell Biology, Genetics and Evolution**)
Name of Scheme:	BS Chemistry (4 Years)
Semester :	3rd
Module Type:	General
Module Rating:	2 Credits

1. **Introduction of the Course:**
This course is planned to provide adequate knowledge about morphology and functioning of cell, cellular organelles and mechanisms of cell division, study of genes and their inheritance patterns and concept of evolution. It is generally aimed to familiarize students with the cell structure and its functioning along with basic concepts of genetics.

2. **Course Objectives:**
The course is designed:
   1. To provide an adequate knowledge about basic concepts of different parts of cells and their morphological/anatomical characteristics along with their functions.
   2. To provide basics of Genetics and Evolution.

"Principles of Physical Chemistry"; Carwan Book House, Lahore.
3. Maron S.H. & B. Jerome, "Fundamentals of Physical Chemistry", Macruthan Publishing Co., Inc. New York. (Also published by National Book Foundation).
4. Atikins P.W.& M.J.Clugston, "Principles of Physical Chemistry" Pitman Publishing Company (1988).
5. Moore W.J. "Physical Chemistry", 5th Ed. Longmans Publishers.
6. Jones M. "Elements of Physical Chemistry" Addison-Sesky Publishing Company.
7. G.M.Barrow, International six Edition "Physical Chemistry".
8. IRA. N. Levine fourth edition "Physical Chemistry"
9. Alberty and Danials, "Physical Chemistry"
10. Castallon, "Physical Chemistry"
11. Laidler & Meiser "Physical Chemistry"
12. Friemental "Chemistry in Action"

Module Code:	Bot - 201
Module title:	Botany – III (**Cell Biology, Genetics and Evolution**)
Name of Scheme:	BS Chemistry (4 Years)
Semester :	3rd
Module Type:	General
Module Rating:	2 Credits

1. **Introduction of the Course:**
This course is planned to provide adequate knowledge about morphology and functioning of cell, cellular organelles and mechanisms of cell division, study of genes and their inheritance patterns and concept of evolution. It is generally aimed to familiarize students with the cell structure and its functioning along with basic concepts of genetics.

2. **Course Objectives:**
The course is designed:
1. To provide an adequate knowledge about basic concepts of different parts of cells and their morphological/anatomical characteristics along with their functions.
2. To provide basics of Genetics and Evolution.

3. **Course Contents**

1. Cell Biology:

1.1. Structures and brief description of biomolecules, carbohydrates, lipids, proteins, nucleic acids.

1.2. Cell: Physico-chemical nature of plasma membrane and cytoplasm.
1.3. Ultrastructure of plant cell with a brief description and functions of the following organelles: Endoplasmic

reticulum, Plastids, Mitochondria, Ribosomes, Dictyosomes, Vacuole, Microbodies (Glyoxysomes and Peroxisomes)
1.4 Nucleus: Nuclear membrane, nucleolus, ultrastructure and morphology of chromosomes, karyotype analysis;
1.5. Reproduction in somatic and embryogenic cell; Mitosis and meiosis; Cell cycle;
1.6. Chromosomal aberrations; Changes in the number of chromosomes, aneuploidy and euploidy; Changes in the structure of chromosomes, deletion, duplication, inversion and translocation, special types of chromosomes, Chromosome systems (parthenogenesis and apomixis).

2. Genetics and Evolution:
2.1. Introduction, scope and brief history of genetics.
2.2. Mendelian inheritance; Laws of segregation and independent assortment, back cross, test cross; Dominance and incomplete dominance.
2.3. Sex linked inheritance, Sex Linkage in *Drosophila* and Man (Color blindness), XO, XY, WZ mechanisms, Sex limited and sex linked characters, Sex determination.
2.4. Linkage and crossing over; Linkage groups, Construction of linkage maps, Detection of linkage.
2.5. Recombination; DNA replication; Nature of gene, Genetic code; Transcription, Translation.
2.6. Regulation of gene expression (*e.g. lac* operon).
2.7. Transmission of genetic material in bacteria; conjugation and gene recombination in co-transduction and transformation;
2.8. Principles of genetic engineering, basic genetic engineering techniques;
2.9. The process and concept of evolution, theories of origin in life, historic idea of evolution, sources of variability, different mechanisms of gene change, role of gene mutation in evolution.

4. **Teaching-learning Strategies**
1. Lectures
2. Group Discussion
3. Laboratory work
4. Seminar/ Workshop

5. **Learning Outcomes:**
1. Students are expected to get familiarized with the morphological and anatomical concepts about different cell parts.
2. They will be able to describe, apply and integrate the basic concepts of Cell Biology including Genetics and Evolution, Biochemistry, Physiology as well as Structure and Functions of different Organelles.
3. This will enable them qualify for basic to moderate level jobs involving general knowledge of Biology.
4. The obtained knowledge shall also enable the students to enter into various entrepreneurial activities involving general introduction to botany.

6. **Assessment Strategies**:
1. Lecture Based Examination (Objective and Subjective)
2. Assignments
3. Class discussion
4. Quiz
5. Tests

7. **<u>Recommended Readings</u>:**
1. Bretscher, A. (2007). *Molecular Cell Biology.* W. H. Freeman and Company
2. Carroll, S. B., Grenier, J. K. and Velnerbee S. D. (2001). From DNA to Diversity—Molecular Genetics and the Evolution of Aminal Design. Blackwell Science.
3. Dyonsager, V. R. (2000). *Cytology and Genetics.* (3rd Ed.), TATA and McGraw Hill Publication Co. Ltd, New Delhi.
4. Gilmartin, P. M. and Bowler, C. (2002). *Molecular Plant Biology.* (Vol. 1 & 2). Oxford University Press. UK.
5. Griffiths, J. F., Miller, J. H., Suzuki, D. T., Lewontin, R. C. and W. M. Gelbart (2010). *An Introduction to Genetic Analysis.* W.H. Freeman and Company.
6. Hoelzel, A. R. (2001). *Conservation Genetics.* Kluwer Academic Publishers.
7. Karp, G. (2002). *Cell and Molecular Biology.* Concepts and Experiments. (4th Ed.), John Wiley and Sons. New York.
8. Lodish, H. (2001). *Molecular Cell Biology.* W.H. Freeman and Company.
9. Sinha, U. and Sinha S. (2003). *Cytogenesis, Plant Breeding and Evolution.* Vini Educational Books, New Delhi.
10. Strickberger, M. V. (2003). *Genetics.* MacMillan Press Ltd., London.
11. Weaver, R. F. (2008). *Molecular Biology.* McGraw Hill, St. Louis.

Module Code:	Math - 211
Module title:	**Elementary Mathematics-II (Calculus)**

7. **Recommended Readings:**

1. Bretscher, A. (2007). *Molecular Cell Biology*. W. H. Freeman and Company
2. Carroll, S. B., Grenier, J. K. and Velnerbee S. D. (2001). From DNA to Diversity—Molecular Genetics and the Evolution of Aminal Design. Blackwell Science.
3. Dyonsager, V. R. (2000). *Cytology and Genetics*. (3rd Ed.), TATA and McGraw Hill Publication Co. Ltd, New Delhi.
4. Gilmartin, P. M. and Bowler, C. (2002). *Molecular Plant Biology*. (Vol. 1 & 2). Oxford University Press. UK.
5. Griffiths, J. F., Miller, J. H., Suzuki, D. T., Lewontin, R. C. and W. M. Gelbart (2010). *An Introduction to Genetic Analysis*. W.H. Freeman and Company.
6. Hoelzel, A. R. (2001). *Conservation Genetics*. Kluwer Academic Publishers.
7. Karp, G. (2002). *Cell and Molecular Biology*. Concepts and Experiments. (4th Ed.), John Wiley and Sons. New York.
8. Lodish, H. (2001). *Molecular Cell Biology*. W.H. Freeman and Company.
9. Sinha, U. and Sinha S. (2003). *Cytogenesis, Plant Breeding and Evolution*. Vini Educational Books, New Delhi.
10. Strickberger, M. V. (2003). *Genetics*. MacMillan Press Ltd., London.
11. Weaver, R. F. (2008). *Molecular Biology*. McGraw Hill, St. Louis.

Module Code:	Math - 211
Module title:	**Elementary Mathematics-II (Calculus)**
Name of Scheme:	BS Chemistry (4 Years)
Semester :	3rd
Module Type:	General
Module Rating:	3 Credits

1. **Introduction of the Course:**
The course provides information about Preliminaries, Limits and Continuity and Derivatives and their Applications.
2. **Course Objectives:**
   1. To prepare the students, not majoring in mathematics, with the essential tools of calculus to apply the concepts and the techniques in their respective disciplines.

3. **Course Outline:**
   1. Preliminaries: Real-number line, functions and their graphs, solution of equations involving absolute values, inequalities.
   2. Limits and Continuity: Limit of a function, left-hand and righthand limits, continuity, continuous functions.
   3. Derivatives and their Applications: Differentiable functions, differentiation of polynomial, rational and transcendental functions, derivatives.
   4. Integration and Definite Integrals: Techniques of evaluating indefinite integrals, integration by substitution, integration by parts, change of variables in indefinite integrals.

4. **Teaching-learning Strategies**
   1. Lectures
   2. Group Discussion
   3. Laboratory work
   4. Seminar/ Workshop

5. **Learning Outcome:**
   1. Students are expected to get familiarized with the with the essential tools of calculus to apply the concepts and the techniques in their respective disciplines

6. **Assessment Strategies:**
   1. Lecture Based Examination (Objective and Subjective)
   2. Assignments
   3. Class discussion
   4. Quiz
   5. Tests

7. **Recommended Readings:**

1. Anton H, Bevens I, Davis S, Calculus: A New Horizon (8 th edition), 2005, John Wiley, New York
2. Stewart J, Calculus (3 rd edition), 1995, Brooks/Cole (suggested text)
3. Swokowski EW, Calculus and Analytic Geometry, 1983, PWS-Kent Company, Boston
4. Thomas GB, Finney AR, Calculus (11 th edition), 2005, Addison-Wesley, Reading, Ma, USA

Module Code:	Zool - 201
Module title:	Zoology – III (**Animal Form And Function- I**)

Name of Scheme:	BS Chemistry (4 Years)
Semester :	3rd
Module Type:	General
Module Rating:	2 Credits

1. **Introduction of the Course:**
The Objectives of the courses are:
   1. To teach about animals' diversity adapted in different strategies' for performance of their similar functions through modifications in body parts in past and present times.
   2. To impart understanding of diverse strategic structural adaptations in each of the functions of integumentary, skeletal, muscular, nervous and sensory, endocrine, circulatory and respiratory systems for effective survival in their specific conditions.
   3. To understand the organ systems, their specialization and coordination with each other and constantly changing internal and external environment, inside and outside the animal's body.
   4. To embrace the phenomena in basic structure of each system that determines its particular function.

2. **Course Objectives:**
The Objectives of the courses are:
   1. To teach about animals' diversity adapted in different strategies' for performance of their similar functions through modifications in body parts in past and present times.
   2. To impart understanding of diverse strategic structural adaptations in each of the functions of integumentary, skeletal, muscular, nervous and sensory, endocrine, circulatory and respiratory systems for effective survival in their specific conditions.
   3. To understand the organ systems, their specialization and coordination with each other and constantly changing internal and external environment, inside and outside the animal's body.
   4. To embrace the phenomena in basic structure of each system that determines its particular function.

3. **Course Outline:**

7. **Recommended Readings:**
   1. Anton H, Bevens I, Davis S, Calculus: A New Horizon (8 th edition), 2005, John Wiley, New York
   2. Stewart J, Calculus (3 rd edition), 1995, Brooks/Cole (suggested text)
   3. Swokowski EW, Calculus and Analytic Geometry, 1983, PWS-Kent Company, Boston
   4. Thomas GB, Finney AR, Calculus (11 th edition), 2005, Addison-Wesley, Reading, Ma, USA

Module Code:	Zool - 201
Module title:	Zoology – III (**Animal Form And Function- I**)

Name of Scheme:	BS Chemistry (4 Years)
Semester :	3rd
Module Type:	General
Module Rating:	2 Credits

1. **Introduction of the Course:**
   The Objectives of the courses are:
   1. To teach about animals' diversity adapted in different strategies' for performance of their similar functions through modifications in body parts in past and present times.
   2. To impart understanding of diverse strategic structural adaptations in each of the functions of integumentary, skeletal, muscular, nervous and sensory, endocrine, circulatory and respiratory systems for effective survival in their specific conditions.
   3. To understand the organ systems, their specialization and coordination with each other and constantly changing internal and external environment, inside and outside the animal's body.
   4. To embrace the phenomena in basic structure of each system that determines its particular function.

2. **Course Objectives:**
   The Objectives of the courses are:
   1. To teach about animals' diversity adapted in different strategies' for performance of their similar functions through modifications in body parts in past and present times.
   2. To impart understanding of diverse strategic structural adaptations in each of the functions of integumentary, skeletal, muscular, nervous and sensory, endocrine, circulatory and respiratory systems for effective survival in their specific conditions.
   3. To understand the organ systems, their specialization and coordination with each other and constantly changing internal and external environment, inside and outside the animal's body.
   4. To embrace the phenomena in basic structure of each system that determines its particular function.

3. **Course Outline:**
   1. Protection, Support, and Movement:
   - Protection: the integumentary system of invertebrates and vertebrates;
   - Movement and support: the skeletal system of invertebrates and vertebrates;
   - Movement: non-muscular movement; an introduction to animal muscles; the muscular system of invertebrates and vertebrates

   2. Communication I:
   - Nerves: Neurons: structure and function.

   3. Communication II:
   - Senses: Sensory reception: baroreceptors,chemoreceptors, georeceptors, hygroreceptors, phonoreceptors, photoreceptors, proprioceptors, tactile receptors, and thermoreceptors of invertebrates
   - Lateral line system and electrical sensing, lateral-line system and mechanoreception, hearing and equilibrium in air and water, skin sensors of mechanical stimuli, sonar, smell, taste and vision in vertebrates.

   4. Communication III:
   - The Endocrine System and Chemical Messengers: Chemical messengers: hormones chemistry; and their feedback systems; mechanisms of hormone action
   - Hormones with principal function each of Porifera, Cnidarians, Platyhelminthes, Nemerteans, Nematodes, Molluscs, Annelids, Arthropods, and Echinoderms invertebrates; an overview of the vertebrate endocrine system; endocrine systems of vertebrates, endocrine systems of birds and mammals

   5. Circulation and Immunity:
   - Internal transport and circulatory systems in invertebrates
   - Characteristics of invertebrate coelomic fluid, hemolymph, and blood cells
   - transport systems in vertebrates; characteristics of vertebrate blood, blood cells and vessels; the hearts and circulatory systems of bony fishes, amphibians, reptiles, birds and mammals; the human heart: blood pressure and the lymphatic system; immunity: nonspecific defenses, the immune response

4. **Teaching-learning Strategies**
   1. Lectures

   2. Group Discussion
   3. Laboratory work
   4. Seminar/ Workshop

5. **Learning Outcome:**
   1. **Acquire** the concept that for the performance of a function for example exchange of respiratory gases the different forms are adapted in the environments e.g. gills in aquatic and lungs in terrestrial environment.
   2. **Understand** that diverse forms adapted to perform the same functions are because of the different past and present conditions.
   3. **Solve** of emergence of diversity of forms for the performance of similar function.
   4. **Analyze** the requirements of diverse forms for the performance of similar function in their past and present needs.
   5. **Evaluate** the adaptations in forms for its efficiency in managing the function in differing situations in the past and present times.
   6. **Demonstrate** that a form is successfully adapted to perform a function adequately and successfully.

6. **Assessment Strategies**:
   1. Lecture Based Examination (Objective and Subjective)
   2. Assignments
   3. Class discussion
   4. Quiz
   5. Tests

7. **Recommended Readings:**

   1. Pechenik, J.A. 2013. Biology of Invertebrates, 4th Ed. (International), Singapore: McGraw-Hill.
   2. Hickman, C.P., Roberts, L.S., Larson, A. 2004. Integrated Principlesof Zoology, 11th Ed. (International), Singapore: McGraw-Hill.
   3. Miller, S.A. and Harley, J.B. 2002. Zoology, 5th Ed. (International), Singapore: McGraw-Hill.
   4. Campbell, N.A. 2002. Biology, 6th Ed. Menlo Park, California:Benjamin/Cummings Publishing
   5. Kent, G.C., Miller, S. 2001. Comparative Anatomy of Vertebrates. NewYork: McGraw-Hill.
   6. Hickman, C.P., Kats, H.L. 2000. Laboratory Studies in IntegratedPrinciples of Zoology. Singapore: McGraw-Hill.

4. **Analyze** the requirements of diverse forms for the performance of similar function in their past and present needs.
5. **Evaluate** the adaptations in forms for its efficiency in managing the function in differing situations in the past and present times.
6. **Demonstrate** that a form is successfully adapted to perform a function adequately and successfully.

6. **Assessment Strategies**:
   1. Lecture Based Examination (Objective and Subjective)
   2. Assignments
   3. Class discussion
   4. Quiz
   5. Tests

7. **Recommended Readings:**

1. Pechenik, J.A. 2013. Biology of Invertebrates, 4th Ed. (International), Singapore: McGraw-Hill.
2. Hickman, C.P., Roberts, L.S., Larson, A. 2004. Integrated Principlesof Zoology, 11th Ed. (International), Singapore: McGraw-Hill.
3. Miller, S.A. and Harley, J.B. 2002. Zoology, 5th Ed. (International), Singapore: McGraw-Hill.
4. Campbell, N.A. 2002. Biology, 6th Ed. Menlo Park, California:Benjamin/Cummings Publishing
5. Kent, G.C., Miller, S. 2001. Comparative Anatomy of Vertebrates. NewYork: McGraw-Hill.
6. Hickman, C.P., Kats, H.L. 2000. Laboratory Studies in IntegratedPrinciples of Zoology. Singapore: McGraw-Hill.

Module Code:	Zool - 202
Module title:	Zoology – III (**Animal Form And Function- I**)
Name of Scheme:	BS Chemistry (4 Years)
Semester :	3rd
Module Type:	General
Module Rating:	1 Credits

1. **Introduction of the Course:**
   The course is organized to provide an adequate knowledge about classification of organisms; definition, concept, evolutionary relationships and tree diagrams; patterns of organization and biodiversity.

2. **Course Objectives:**
   The Objectives of the courses are:
   1. To teach about animals' diversity adapted in different strategies' for performance of their similar functions through modifications in body parts in past and present times.
   2. To impart understanding of diverse strategic structural adaptations in each of the functions of integumentary, skeletal, muscular, nervous and sensory, endocrine, circulatory and respiratory systems for effective survival in their specific conditions.
   3. To understand the organ systems, their specialization and coordination with each other and constantly changing internal and external environment, inside and outside the animal's body.
   4. To embrace the phenomena in basic structure of each system that determines its particular function.

3. **Course Contents**

Practicals:
1. Study of insect chitin, fish scale, amphibian skin, reptilian scales,feathers and mammalian skin.
2. Study and notes of skeleton of Labeo (*Labeo rohita)*, Frog (*Hoplobatrachus tigerinus*), Varanus (*Varanus bengalensis)*, fowl*(Gallus gallus domesticus)* andrabbit (*Oryctolagus cuniculus).*

Note: Exercises of notes on the adaptations of skeletons to their function must be done.

3. Earthworm or leech; cockroach, freshwater mussel, Channa or Catla catla or Labeo or any other local fish, frog, pigeon and rat or mouse and rabbits dissections as per availability.
4. Study of heart, principal arteries and veins in a representative vertebrate (dissection of representative fish/mammals).

4. **Teaching-learning Strategies**
   1. Lectures
   2. Group Discussion
   3. Laboratory work
   4. Seminar/ Workshop

5. **Learning Outcome:**
   1. **Acquire** the concept that for the performance of a function for example exchange of respiratory gases the different forms are adapted in the environments e.g. gills in aquatic and lungs in terrestrial environment.
   2. **Understand** that diverse forms adapted to perform the same functions are because of the different past and present conditions.
   3. **Solve** of emergence of diversity of forms for the performance of similar function.
   4. **Analyze** the requirements of diverse forms for the performance of similar function in their past and present needs.
   5. **Evaluate** the adaptations in forms for its efficiency in managing the function in differing situations in the past and present times.
   6. **Demonstrate** that a form is successfully adapted to perform a function adequately and successfully.

6. **Assessment Strategies**:
   1. Lecture Based Examination (Objective and Subjective)
   2. Assignments
   3. Class discussion
   4. Quiz
   5. Tests

7. **Books Recommended:**

1. Pechenik, J.A. 2013. Biology of Invertebrates, 4th Ed. (International), Singapore: McGraw-Hill.
2. Hickman, C.P., Roberts, L.S., Larson, A. 2004. Integrated Principlesof Zoology, 11th Ed. (International), Singapore: McGraw-Hill.
3. Miller, S.A. and Harley, J.B. 2002. Zoology, 5th Ed. (International), Singapore: McGraw-Hill.
4. Campbell, N.A. 2002. Biology, 6th Ed. Menlo Park, California:Benjamin/Cummings Publishing
5. Kent, G.C., Miller, S. 2001. Comparative Anatomy of Vertebrates. NewYork: McGraw-Hill.
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Semester - IV

Module Code:	Eng – 202
Module title:	English IV (English for Practical Aims)
Name of Scheme:	BS Chemistry (4 Years)
Semester :	4th
Module Type:	Compulsory
Module Rating:	3 Credits

1. **Introduction of the course:**